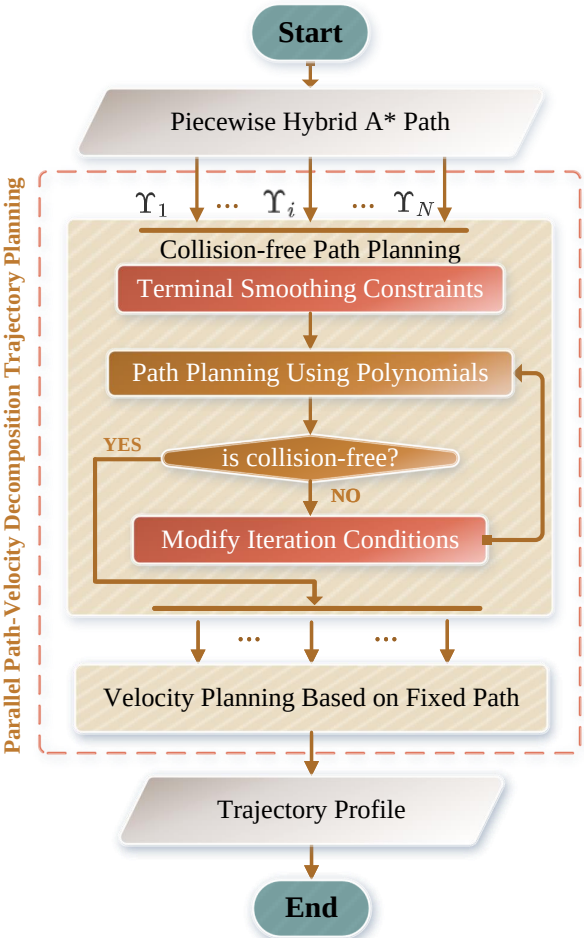
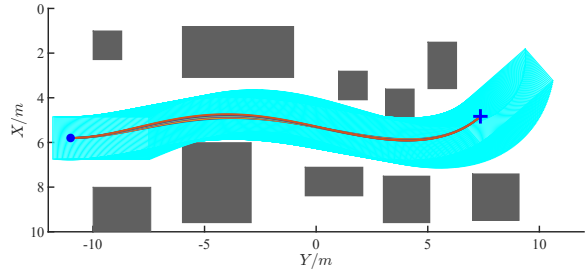
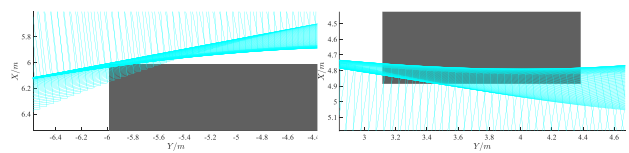


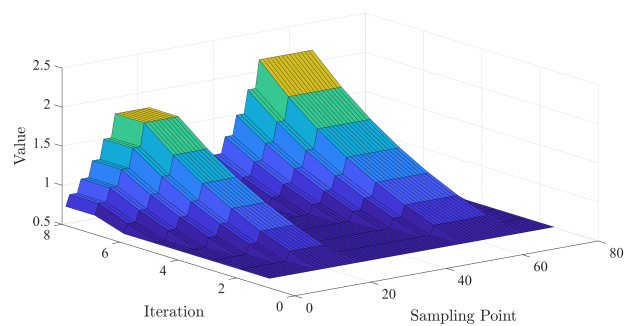
Flowchart of the Rapid Iterative Trajectory Planning (RITP) Method. The proposed method effectively balances the necessity for time efficiency and precise collision avoidance with a novel collision avoidance framework. Moreover, it enhances the overall control fea-sibility of the planned trajectory by incorporating the vehicle kinemat-ics model and including terminal smoothing constraints (TSC) at gear shifting points (GSP) during path planning.



(a) The cyan path indicates the planning result at the first iteration. The different colored curves represent the planned paths generated during iterations.



(b) Details of the collision between the planned path and obsta-cles in the first iteration.



(c) Schematic diagram of the variation of the error vector **e** dur-ing Iterative Collision Avoidance (ItCA) process.

Schematic illustration of the effectiveness of the ItCA method. In the first iteration, the planned path collides with obsta-cles, and after a finite number of iterations, ItCA path planning yields a collision-free path. The two peaks indicate the highest values of the error vector **e** corresponding to the two collision parts of the iterative path.